

$$\begin{aligned}
 P_n(t) = & \\
 & (1-\rho)\rho^n Q(\sqrt{2\mu t}, \sqrt{2\lambda t}) + \\
 & \exp(-(\lambda+\mu)t) \left\{ \rho^{(n-i)/2} I_{n-i}(2t\sqrt{\lambda\mu}) + \rho^{(n-i-1)/2} I_{n+i+1}(2t\sqrt{\lambda\mu}) - \right. \\
 & \left. (1-\rho)\rho^n \sum_{l=0}^{n+i+1} \rho^{-l/2} I_l(2t\sqrt{\lambda\mu}) \right\}. \quad (2.182)
 \end{aligned}$$

Computational results reported in [JONE] indicate that this expression is up to seven times faster to evaluate than the original expression with the infinite summation. Some additional references dealing with computation in the transient case appear in To Look Further at the end of this chapter.

2.12 The M/G/1 Queueing System

2.12.1 Introduction

We have been able to get some very specific results throughout this chapter by assuming Poisson arrivals and an exponential server(s). This has been due to the memoryless nature of these models. Surprisingly, some very simple and powerful results can be derived for the M/G/1 queueing system where one assumes that arrivals are again Poisson but that service times are described by an arbitrary ("general") probability distribution. These are the famous Pollaczek-Khinchin formulas [POLL], [KHIN]. The first one at which we will look is for the mean number of customers in the queueing system. The second one at which we will look is a transform equation for the state probabilities.

2.12.2 Mean Number in the Queueing System

Introduction

Our basic approach will be to write a difference equation for the number in the queueing system immediately after the i th departure. By taking the expectation of the square of this expression we will be able to develop the Pollaczek-Khinchin mean value formula. A key point is that the state probabilities at these departure

instants are in fact equal to the state probabilities at any time instant. This is something that will be assumed true for now but is proved explicitly in section 12.3. This approach that we take was first described by David Kendall in 1951 [KEND]. However, the actual mean value expression goes back to Pollaczek in 1930 [POLL] and Khinchin in 1932 [KHIN] where it was found by means of more complex derivations. As many other authors have done ([GROS], [HAMM], [KLEI 75]) we will expand on Kendall's approach.

The Recursion

Let us start. We will look at the M/G/1 queueing system at the instants during which departures occur. The number in the system *immediately after* the $(i+1)$ st departure instant is n_{i+1} . This number includes both those customers in the queue and the customer in service. This number is also equal to the number in the system immediately after the i th departure instant minus 1 (because of the departure of a customer at the $(i+1)$ st departure instant) plus the number that arrive into the system between the i th and $(i+1)$ st departure instants. This number of arrivals we will call a_{i+1} . For this equation to work there must be at least one customer in the system at the i th departure instant. Thus

$$n_{i+1} = n_i - 1 + a_{i+1}, \quad n_i > 0. \quad (2.183)$$

Of course, we still have to take care of the case $n_i = 0$. This is actually simpler than the case $n_i > 0$. The number in the queueing system at the $(i+1)$ st departure instant is just the number of arrivals during the service period of the first customer to arrive to the empty queueing system. This first customer is the one that departs at the $(i+1)$ st departure instant and is not counted in determining n_{i+1} . So

$$n_{i+1} = a_{i+1}, \quad n_i = 0. \quad (2.184)$$

Putting this together we have

$$n_{i+1} = \begin{cases} n_i - 1 + a_{i+1}, & n_i > 0, \\ a_{i+1}, & n_i = 0. \end{cases} \quad (2.185)$$

We would like to express this as a single equation. To do this, we will introduce the unit step function:

$$u(n_i) = \begin{cases} 1, & n_i > 0, \\ 0, & n_i = 0. \end{cases} \quad (2.186)$$

Now we can rewrite the equations for n_{i+1} as

$$n_{i+1} = n_i - u(n_i) + a_{i+1}. \quad (2.187)$$

If we square both sides of this recursion and take expectations we have

$$E[(n_{i+1})^2] = E[(n_i - u(n_i) + a_{i+1})^2], \quad (2.188)$$

$$\begin{aligned} E[(n_{i+1})^2] = & \quad (2.189) \\ & E[n_i^2] + E[(u(n_i))^2] + E[a_{i+1}^2] \\ & - 2E[n_i u(n_i)] + 2E[n_i a_{i+1}] - 2E[u(n_i) a_{i+1}]. \end{aligned}$$

In order to solve for the Pollaczek-Khinchin formula we need to solve for each of these terms, one by one. We will now do this

Sundry Results

$$E[(n_{i+1})^2], E[n_i^2]:$$

These two terms should be equal in equilibrium so that they will cancel in (2.189).

$$E[(u(n_i))^2]:$$

Because of the nature of the step function $u(n_i)^2 = u(n_i)$, we can replace $E[(u(n_i))^2]$ with $E[u(n_i)]$. But what is $E[u(n_i)]$? We will have to digress in order to solve for it. Now

$$E[u(n_i)] = \sum_{n=0}^{\infty} u(n_i) P[n_i = n], \quad (2.190)$$

$$E[u(n_i)] = \sum_{n=1}^{\infty} P[n_i = n], \quad (2.191)$$

$$E[u(n_i)] = P[busy] \quad (2.192)$$

Luckily, one can easily determine $P[busy]$ for any queueing system. An argument appearing in [KLEI 75] will be used. Consider an interval I . The server will be busy for a time period equal to

$$I - Ip_0.$$

The number of customers served is

$$(I - Ip_0)\mu$$

where μ is the mean service rate of the queueing system. In equilibrium the quantity above should be equal to the number of arrivals:

$$\lambda I = (I - Ip_0)\mu. \quad (2.193)$$

Rearranging this one has

$$\rho = 1 - p_0. \quad (2.194)$$

Note that no restrictive assumptions have been made about either the arrival process or the service times. Thus this result is valid for a G/G/1 queueing system, that is, one with an arbitrary arrival process and service time distribution.

We can now bring all this together to say that

$$E[(u(n_i))^2] = E[u(n_i)] = P[busy] = 1 - p_0 = \rho. \quad (2.195)$$

$$2E[n_i u(n_i)]:$$

Clearly $n_i u(n_i) = n_i$, so we can say

$$2E[n_i u(n_i)] = 2E[n_i]. \quad (2.196)$$

$$2E[u(n_i)a_{i+1}]:$$

The number of customers that arrive into the system between the i th and $(i+1)$ st departure instant, a_{i+1} , is independent of the number in the queueing system immediately after the i th departure instant, n_i . Thus

$$E[u(n_i)a_{i+1}] = E[u(n_i)]E[a_{i+1}]. \quad (2.197)$$

We already know that $E[u(n_i)] = \rho$. To determine $E[a_{i+1}]$ one can look at the original recursion for $E[n_{i+1}]$

$$n_{i+1} = n_i - u(n_i) + a_{i+1} \quad (2.198)$$

and take the expectation of both sides:

$$E[n_{i+1}] = E[n_i] - E[u(n_i)] + E[a_{i+1}]. \quad (2.199)$$

In equilibrium $E[n_{i+1}] = E[n_i]$, so one is left with

$$E[a_{i+1}] = E[u(n_i)] = \rho. \quad (2.200)$$

One can now say that:

$$2E[u(n_i)a_{i+1}] = 2E[u(n_i)]E[a_{i+1}] = 2\rho^2. \quad (2.201)$$

$$2E[n_i a_{i+1}]:$$

Using the same argument as before, the two quantities are statistically independent, so

$$2E[n_i a_{i+1}] = 2E[n_i]E[a_{i+1}] = 2E[n_i]\rho. \quad (2.202)$$

Putting It All Together

Now we can substitute these sundry results into the equation:

$$E[(n_{i+1})^2] = \quad (2.203)$$

$$E[n_i^2] + E[(u(n_i))^2] + E[a_{i+1}^2]$$

$$- 2E[n_i u(n_i)] + 2E[n_i a_{i+1}] - 2E[u(n_i) a_{i+1}].$$

This results in

$$0 = 0 + \rho + E[a_{i+1}^2] - 2E[n_i] + 2E[n_i]\rho - 2\rho^2. \quad (2.204)$$

Solving for $E[n_i]$ we have

$$E[n_i] = \frac{\rho + E[a_{i+1}^2] - 2\rho^2}{2(1 - \rho)}. \quad (2.205)$$

We will assume for now, and prove in section 2.12.3, that the expected number of customers in the queueing system at the departure instants is the same as the expected number of customers at *any* instant of time:

$$E[n_i] = E[n]. \quad (2.206)$$

We can also say, because the input is Poisson, that the expected number of arrivals between two successive departure instants is a time-invariant quantity:

$$E[a_{i+1}^2] = E[a^2]. \quad (2.207)$$

We thus have

$$E[n] = \frac{\rho + E[a^2] - 2\rho^2}{2(1 - \rho)}. \quad (2.208)$$

We almost have a simple expression for the mean number in the M/G/1 queueing system in equilibrium or $E[n]$. What is lacking is a simple expression for $E[a^2]$

What is $E[a^2]$?

The quantity "a" is the number of arrivals during a typical servicing period in equilibrium. Proceeding as in [GROS] we can write

$$E[a^2] = \text{VAR}[a] + (E[a])^2, \quad (2.209)$$

$$E[a^2] = \text{VAR}[a] + \rho^2. \quad (2.210)$$

We need to solve for $\text{VAR}[a]$. Let s be the service time. Then an interesting relationship [GROS] for two random variables X and Y that we can make use of is [PARZ]

$$\text{VAR}[Y] = E[\text{VAR}(Y|X)] + \text{VAR}[E(Y|X)]. \quad (2.211)$$

In terms of our application this becomes

$$\text{VAR}[a] = E[\text{VAR}(a|s)] + \text{VAR}[E(a|s)]. \quad (2.212)$$

One can solve for the two components of this sum:

$$E[\text{VAR}(a|s)] = E[\lambda s] = \lambda E[s] = \rho, \quad (2.213)$$

$$\text{VAR}[E(a|s)] = \text{VAR}[\lambda s] = \lambda^2 \sigma_s^2. \quad (2.214)$$

Here σ_s^2 is the variance of the service time. So we have

$$\text{VAR}[a] = \rho + \lambda^2 \sigma_s^2 \quad (2.215)$$

and

$$E[a^2] = \rho + \lambda^2 \sigma_s^2 + \rho^2. \quad (2.216)$$

Pollaczek-Khinchin Mean Value Formula

From before we have

$$E[n] = \frac{\rho + E[a^2] - 2\rho^2}{2(1-\rho)}. \quad (2.217)$$

Substituting (2.216) for $E[a^2]$ leads to

$$E[n] = \frac{\rho + \rho + \lambda^2 \sigma_s^2 + \rho^2 - 2\rho^2}{2(1-\rho)} \quad (2.218)$$

$$E[n] = \frac{2\rho - \rho^2 + \lambda^2 \sigma_s^2}{2(1-\rho)}. \quad (2.219)$$

This is one form of the Pollaczek-Khinchin mean value formula. We can express it in another form by expressing $E[n]$ as

$$E[n] = \frac{2\rho - 2\rho^2 + \rho^2 + \lambda^2 \sigma_s^2}{2(1-\rho)} \quad (2.220)$$

$$E[n] = \rho + \frac{\rho^2 + \lambda^2 \sigma_s^2}{2(1-\rho)}. \quad (2.221)$$

What is amazing about the Pollaczek-Khinchin mean value formula is that it expresses $E[n]$ as a function of the mean arrival rate, mean service rate, and variance of the service time distribution. One might think that since the service time distribution is general, the formula should include higher moments of the service time distribution. But this is not the case. The formula depends only on the mean and variance of the service time distribution.

We note in passing that Little's Law can be used in conjunction with this mean value formula to develop expressions for the mean waiting time, $\bar{\tau}$.

Example: In the M/D/1 queueing system the arrivals are Poisson and the service time is a fixed, deterministic quantity. This is actually a good model for the processing of fixed length packets in a communications network.

Let us make a comparison. For the M/M/1 queueing system the Pollaczek-Khinchin mean value formula reduces to the result we had previously:

$$E[n] = \frac{\rho}{1 - \rho}.$$

For the M/D/1 queueing system $\sigma_s^2 = 0$, so

$$E[n] = \rho + \frac{\rho^2}{2(1 - \rho)}.$$

Table 2.5 compares $E[n]$ for both types of queueing systems for various values of ρ

Table 2.5: $E[n]$		
ρ	M/M/1	M/D/1
0.1	0.111	0.106
0.2	0.250	0.225
0.3	0.429	0.364
0.4	0.667	0.533
0.5	1.00	0.750
0.6	1.50	1.05
0.7	2.33	1.52
0.8	4.00	2.40
0.9	9.00	4.95
0.99	99.0	50.0

Note that the expected number in Table 2.5 is smaller for the M/D/1 queueing system. In fact, we can show, with some algebraic manipulation, that for the M/D/1 queueing system

$$E[n] = \frac{\rho}{1 - \rho} - \frac{\rho^2}{2(1 - \rho)}.$$

The $E[n]$ is equal to that for the M/M/1 queueing system, less a positive quantity. Thus $E[n]$ is always smaller for an M/D/1 queueing system compared to an M/M/1 queueing system.

It is also straightforward to show, as Table 2.5 suggests, that as $\rho \rightarrow 1$ the expected number of customers in an M/M/1 queueing system is twice that in an M/D/1 queueing system.

2.12. The M/G/1 Queueing System

Although the M/D/1 queueing system is a good model for packet processing, the problem is that we only have simple analytical results for a single such system. Because networks with Poisson arrivals and deterministic service times are analytically intractable, the memoryless M/M/1 queueing system is often used as an approximation in its place.

▽

2.12.3 Why We Use Departure Instants

An essential part of the previous derivation of the Pollaczek-Khinchin mean value formula is our assumption that the state probabilities at departure instants are the same as the state probabilities at any instant. Following the method used in [GROS 85], we will show that this is true.

We will take a realization of the queueing process over a long interval. The number in the queueing system at time t will be $n(t)$. The interval will stretch from time 0 to time T .

The number of arrivals in the interval $(0, T)$ that occur when the number of customers in the system is n will be called $a_n(T)$. The number of departures bringing the system from state $n+1$ to state n in the interval $(0, T)$ will be called $d_n(T)$.

If one thinks about the movement of the system state in the usual one-dimensional state transition diagram for a birth-death system, one can see that

$$|a_n(T) - d_n(T)| \leq 1. \quad (2.222)$$

It should also be true from first principles that

$$d(T) = a(T) + n(0) - n(T) \quad (2.223)$$

where $d(T)$ and $a(T)$ are the total number of departures and arrivals, respectively, over $(0, T)$. One can now define the state probabilities for the departure instants as

$$\lim_{T \rightarrow \infty} \frac{d_n(T)}{d(T)}, \quad n=1, 2, 3, \dots \quad (2.224)$$

This empirical definition of the state probability becomes exact in the limit. Now

$$\frac{d_n(T)}{d(T)} = \frac{a_n(T) + d_n(T) - a_n(T)}{a(T) + n(0) - n(T)}. \quad (2.225)$$

The denominator can be recognized as the equation we just discussed (2.223). The numerator was created by adding and subtracting $a_n(T)$ to $d_n(T)$. As $T \rightarrow \infty$, $d_n(T) - a_n(T)$ will never be greater than one, as has been mentioned. Also, because we have a stationary process if $n(0)$ is finite, so will be $n(T)$. Thus as $T \rightarrow \infty$, $a(T)$ and $a_n(T)$ will $\rightarrow \infty$, and

$$\lim_{T \rightarrow \infty} \frac{d_n(T)}{d(T)} = \lim_{T \rightarrow \infty} \frac{a_n(T)}{a(T)}. \quad (2.225)$$

What this proves directly is that the state probabilities seen at the departure instants are the same as the state probabilities seen at the arrival instants. But going back to our Poisson process coin-flipping analogy, the arrival instants are random points in time and independent of the queueing system state. Thus it should seem reasonable to say that the state probabilities at arrival instants are the same as the state probabilities seen at any time. This completes the proof that looking at state probabilities at departure instants for the M/G/1 queueing system is the same as looking at state probabilities for this queueing system at any instant.

▽

In Kendall's original 1951 paper [KEND] he called the instants of departures "regeneration points". As he put it: "An epoch is a point of regeneration if a knowledge of the state of the process at that particular epoch has the characteristic Markovian consequence that a statement of the past history of the process loses all its predictive value." He went on further to say that "A Markov process, therefore, is precisely a process for which 'every' epoch is a point of regeneration."

In the printed discussion that followed this paper D. V. Lindley formally expressed a regeneration point as a point, t_0 , where for all $t > t_0$:

$$DIST\{X(t) | X(t_0)\} \equiv DIST\{X(t) | X(\tau) \text{ for all } \tau \leq t_0\}.$$

There is more than one set of regeneration points. Kendall used the set of departure instants in his paper. In the subsequent discussion Lindley suggested the instants when the queueing system becomes empty. He also mentioned the use of the time instants when a customer had been in service x seconds. For simplicity of solution though, the choice of departure instants is probably best.

2.12.4 Probability Distribution of the Number in the Queueing System

What does the Markov chain imbedded at the times of departure look like? Let us define a matrix of transition probabilities as

$$\mathbf{P} = [P_{rs}] \equiv P[n_{i+1}=s | n_i=r]. \quad (2.227)$$

Here n_i is the number in an M/G/1 queueing system immediately after the i th departure. To get from $n_i=r$ to $n_{i+1}=s$ there must be $s-r+1$ arrivals between the i th and the $(i+1)$ st departure instants. The "+1" in $s-r+1$ is necessary because the number in the queueing system is reduced by one due to the departure at the $(i+1)$ st departure instant. Let us set

$$[P_{rs}] = k_{s-r+1} = k_{\# \text{ arrivals}} \quad (2.228)$$

where "# arrivals" is the number of customers that arrive between the i th and the $(i+1)$ st departure instants. We are using k 's because this is the original notation appearing in Kendall's 1951 paper. What we now have is a state transition matrix that looks like Table 2.6:

Table 2.6: $\mathbf{P} = [p_{rs}] =$						
	0	1	2	3	4	·
0	k_0	k_1	k_2	k_3	k_4	·
1	k_0	k_1	k_2	k_3	k_4	·
2	0	k_0	k_1	k_2	k_3	·
3	0	0	k_0	k_1	k_2	·
4	0	0	0	k_0	k_1	·
·	·	·	·	·	·	·

The sum of the k 's in each row must be 1 as this covers the probability of going from state r to *some* (and possibly the same) state. The matrix is also of infinite dimension, but this will not cause any problems in what follows.

How was the placement of matrix entries chosen? In the top row one goes from 0 customers in the queueing system at the i th departure instant to 0,1,2,3,... customers at the $(i+1)$ st departure instant. Thus we need $k_0, k_1, k_2, k_3, \dots$ arrivals. In the second row, for instance, to go from 1 customer at the i th departure instant to 3 customers at the $(i+1)$ st departure instant there must be 3 arrivals during this time. That is, we start with 1 customer at the end of the i th departure instant, three more customers arrive for a total of 4 customers, and one customer leaves at the end of the $(i+1)$ st departure instant for a final total of 3 customers.

We can express k_j , the probability of j arrivals between the i th and the $(i+1)$ st departure instants, explicitly as

$$k_j = \int_0^{\infty} \text{Prob}[j \text{ arrivals} | \text{time } s] b(s) ds$$

(2.230)

where $b(s)$ is the service time density. The variable s is the time between the i th and the $(i+1)$ st departure instants. This time is a constant in equilibrium. Now since the arrival stream is Poisson, we can substitute the Poisson distribution into the integral for

$$k_j = \int_0^{\infty} \frac{(\lambda s)^j e^{-\lambda s}}{j!} b(s) ds. \quad (2.230)$$

The basic idea is that for a known service time density, $b(s)$, one can plug it into the above equation and perform the integration to solve for k_j . We began this section by asking what the Markov chain imbedded at the times of departures would look like. The answer is illustrated in Figure 2.26. Here are shown all the transitions leaving the n th state. The k_j associated with each transition is the probability that one leaves the state via that transition as opposed to another transition at the i th departure instant. It is *not* a transition rate, as in the previous state transition diagrams.

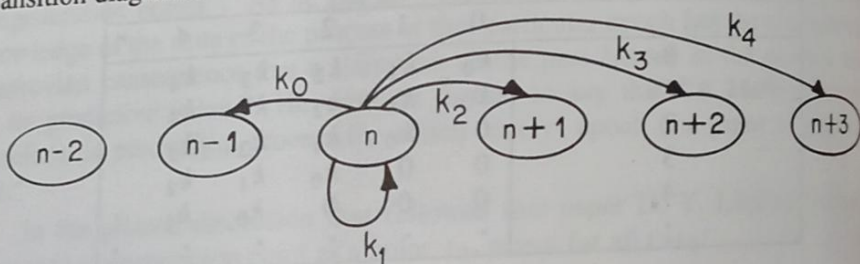


Fig. 2.26: Imbedded Markov Chain for M/G/1

Let us now ask a second question. Now that we have this Markov chain, can we calculate the probability of there being j customers in the queueing system at the departure instants? The answer is yes. We will use an approach described by [GROS]. An alternate approach appears in [KLEI].

We know from the previous section that the state probabilities at these instants are in fact the same as the state probabilities at any instant. The first step is to write the equilibrium equations

$$\pi = \pi P$$

(2.231)

where

$$\pi = [\pi_0 \ \pi_1 \ \pi_2 \ \pi_3 \ \cdots] \quad (2.232)$$

are the state probabilities. We know the structure of P . Equation (2.231) gives rise to equations of the form:

$$\pi_0 = \pi_0 k_0 + \pi_1 k_0, \quad (2.233)$$

$$\pi_1 = \pi_0 k_1 + \pi_1 k_1 + \pi_2 k_0,$$

$$\pi_2 = \pi_0 k_2 + \pi_1 k_2 + \pi_2 k_1 + \pi_3 k_0,$$

$$\pi_3 = \pi_0 k_3 + \pi_1 k_3 + \pi_2 k_2 + \pi_3 k_1 + \pi_4 k_0.$$

These equations can be compactly summarized, where r and s are integers, as:

$$\pi_r = \pi_0 k_r + \sum_{s=1}^{r+1} \pi_s k_{r-s+1}, \quad r=0, 1, 2, 3, \dots \quad (2.234)$$

The second step is to make use of transforms. First we introduce the change of variables $t=s-1$ or $s=t+1$. This gives us:

$$\pi_r = \pi_0 k_r + \sum_{t=0}^r \pi_{t+1} k_{r-t}. \quad (2.235)$$

We will also introduce the transform of the state probabilities

$$\Pi(z) = \sum_{i=0}^{\infty} \pi_i z^i \quad (2.236)$$

and the transform of the k_j 's

$$K(z) = \sum_{i=0}^{\infty} k_i z^i. \quad (2.237)$$

The k 's, after all, are a sequence of real (positive) numbers like any sequence so there is no reason why we cannot take their transform. Finally, the summation in the above difference equation for π_r can be seen to be "almost" convolution. That is, it is similar to

$$x_i = \sum_{j=0}^{\infty} f_j g_{i-j} \quad (2.21)$$

for which we know that

$$X(z) = F(z)G(z) \quad (2.22)$$

where $X(z)$, $F(z)$, and $G(z)$ are the one sided ($i=0$ to ∞) transforms of x_i , f_i , and g_i respectively.

Let us now take the one-sided transform of the previous difference equation for π_r . This is

$$\Pi(z) = \pi_0 K(z) + Z\{\pi_{r+1}\}K(z). \quad (2.23)$$

We can take the transform of the convolution, in spite of the finite index, because π_i and k_i are only defined for $i \geq 0$. In the preceding equation

$$Z\{\pi_{r+1}\} = \sum_{t=0}^{\infty} \pi_{t+1} z^t = z^{-1} \sum_{t=0}^{\infty} \pi_{t+1} z^{t+1}, \quad (2.24)$$

$$Z\{\pi_{r+1}\} = z^{-1} \sum_{t=-1}^{\infty} \pi_{t+1} z^{t+1} - z^{-1} \pi_0, \quad (2.25)$$

$$Z\{\pi_{r+1}\} = z^{-1} \Pi(z) - z^{-1} \pi_0. \quad (2.26)$$

So substituting this into the transform equation yields

$$\Pi(z) = \pi_0 K(z) + [z^{-1} \Pi(z) - z^{-1} \pi_0] K(z) \quad (2.27)$$

or

$$\Pi(z)(1 - z^{-1} K(z)) = \pi_0 (1 - z^{-1}) K(z). \quad (2.28)$$

Solving for $\Pi(z)$ gives us

$$\Pi(z) = \frac{\pi_0(1 - z^{-1})K(z)}{1 - z^{-1}K(z)}. \quad (2.246)$$

Multiplying by $-z/z$ results in

$$\Pi(z) = \frac{\pi_0(1 - z)K(z)}{K(z) - z}. \quad (2.247)$$

To solve for π_0 one can make use of the fact that $\Pi(1)=1$, $K(1)=1$ and $K'(1)=\lambda/\mu$ and use of L'Hopital's rule to obtain $\pi_0 = 1 - \lambda/\mu = 1 - \rho$, so

$$\Pi(z) = \frac{(1-\rho)(1-z)K(z)}{K(z) - z}. \quad (2.248)$$

This is the result that we are seeking. It expresses the transform of the probabilities of the number in the M/G/1 queueing system as a function of the transform of the k_j 's. Recall that we obtain the k_j 's from an integral equation which includes $b(s)$, the service time distribution.

2.13 Priority Systems for Multiclass Traffic

Often, customers can be divided into different classes. For instance, in a computer network there may be a natural distinction between data, voice, and video packets. In a computer system different jobs may be distinguished by their needs for service. In such cases as these it is common to implement service disciplines at the queues that treat customers according to their "priority".

Establishing a priority structure is a natural way to provide different levels of service to different classes of traffic. Each class of traffic then receives service based on its priority and the presence or absence of traffic of higher priority. It is well known that networks with customer priorities do not admit a tractable product form solution (see chapter 3). This can be explained in two ways. One is that a priority system introduces dependencies between customers in a node that is at odds with the necessary independence of customers for a product form solution. The other is that the underlying Markov chains do not have the building block structure which is characteristic of product form networks (see chapter 3). Nonetheless, there has been a growing interest in priority systems for multiclass traffic [KLEI 75], [KIM], [CHIP]. The following discussion provides a summary of some important priority results.